Systematic robustness of a boosted top-quark tagger on ATLAS open data: a constituent-level feature reconstruction, paired SHAP-Delta attribution drift, and conformal coverage

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Abstract

The ATLAS top-tagging open dataset releases detector-systematic and generator-modelling variations as dedicated simulated samples rather than as per-event weights. For the four exactly pairable calorimeter variations, the fifteen stored high-level substructure quantities remain nominal, so we recompute them from the released constituents with a single implementation. Against the stored nominal branches the median relative agreement is $\mathbf{2.5 \times 10^{-8}}$ to $\mathbf{4.8 \times 10^{-7}}$ per quantity on 14,103 jets; the **0.072%** of comparisons beyond a $\mathbf{10^{-3}}$ tolerance fall on 44 jets, none of them a formula or implementation mismatch. The frozen reconstruction was applied to 80,481,421 jet entries. An XGBoost tagger developed from a five-million-jet nominal subset reaches an unweighted area under the receiver-operating-characteristic curve of **0.9320** and background rejection $\mathbf{41.91 \pm 0.12}$ (stat.) at **50%** signal efficiency on 10,306,812 test jets. Its total relative systematic uncertainty is $\mathbf{(25.8 \pm 0.6_{stat} \pm 0.5_{seed})\%}$, dominated by generator modelling (**20.0%**). Under the calorimeter energy-scale and energy-resolution variations, paired TreeSHAP attribution drift is led by $\mathbf{ECF_3}$, the input of highest degree in the constituent momenta, while global importance orderings change minimally (Spearman $\boldsymbol{\rho \geq 0.986}$). Restricting the tagger to dimensionless inputs lowers the six-seed mean total uncertainty from **26.3%** to **15.8%** but background rejection from **41.8** to **18.9**. Class-conditional split-conformal coverage, calibrated once on nominal simulation, stays within one percentage point of the **90%** target on all nine detector-systematic samples.

Keywords: jet substructure, boosted top tagging, machine learning, SHAP, conformal prediction, systematic uncertainties, ATLAS open data

1 Introduction

Identifying boosted top quarks from their hadronic decays is among the harder jet-tagging problems at the Large Hadron Collider [1]. When a top quark is produced with transverse momentum well above its rest mass of about 173 GeV [2], the three quarks from $t \to Wb \to qqb$ merge into a single large-radius jet whose interior carries a three-pronged radiation pattern. Separating such jets from the far more abundant QCD multijet background matters both for measurements of boosted top-pair production and for searches involving top quarks.

Early taggers cut on individual substructure observables such as N-subjettiness ratios [3, 4] or energy correlation functions [5]. Multivariate combinations followed, and deep networks reading low-level constituent information pushed background rejection substantially higher [6–8]; Refs. [9, 10] review that progression. ATLAS and CMS have both published systematic studies of boosted-top classifiers on full detector simulation [11, 12]. The most recent ATLAS study [11] benchmarked several architectures and released the underlying simulation as an open dataset [13], together with a prescribed procedure for turning tagger performance into a systematic uncertainty.

That release is unusual in one respect that shapes this paper. Detector systematic variations and generator-modelling alternatives are distributed as *separate simulated samples*, not as per-event weights. A weight-based analysis of tagger robustness is therefore not available on this dataset. In the four calorimeter variations that pair exactly, all fifteen stored high-level quantities remain nominal despite changed constituents (Section 2.3). A study of how a classifier responds to these variations must first rebuild the feature space from the released constituents.

The contributions of this work are the following.

1. *A reconstructed feature space.* All fifteen high-level quantities are rebuilt from the released constituents with one implementation applied identically on both sides of every comparison, validated against the stored nominal branches, and applied to 80.5 million jet entries (Section 2). This is the enabling step for everything that follows.
2. *A benchmarked tagger with measured precision.* An XGBoost tagger [14, 15] is trained on that feature space and evaluated exactly as ATLAS prescribes for this dataset, so that its performance and systematic uncertainty are directly comparable to the published benchmark. The statistical uncertainty of the nominal performance and of every systematic uncertainty is measured with a bootstrap that preserves the jet-by-jet pairing of the calorimeter variations, and its dependence on the training seed with six independent fits (Section 4.2).
3. *Paired attribution drift.* A *paired* analysis of Shapley additive explanations (SHAP) [16, 17] measures how the attribution assigned to each feature moves for the *same physical jet* under a systematic variation.
4. *An explanation in terms of the inputs.* The leading attribution drift and the dominant detector and modelling uncertainties are traced to the degree of each input in the constituent momenta, by measuring the per-jet response of every input and by retraining on dimensionless inputs (Section 4.3).
5. *Coverage under systematic variation.* A class-conditional split-conformal coverage check [18, 19] asks whether prediction sets calibrated once on nominal simulation retain their nominal coverage on varied simulation.

The two diagnostics operate at different levels, and it is worth being explicit about that from the start. The SHAP analysis is genuinely paired: it is restricted to the four calorimeter-cluster variations whose jets match the nominal sample one for one on an exact identity key, so each attribution difference is a difference for one physical jet. The conformal analysis is not paired. It is a dataset-level comparison in which one fixed pair of thresholds, calibrated once on held-out nominal simulation, is applied unchanged to the nominal test sample and to all nine detector-systematic samples, including the four track samples that are different jet populations altogether. Neither technique is new in itself: tagger robustness under detector systematics [11, 12, 20], Shapley attributions for boosted-top classifiers [20, 21], and conformal prediction in high-energy physics [22] have all been reported before. What is new here is that the attribution comparison is made on physically matched jet pairs drawn from dedicated systematic samples, on a feature space rebuilt so that the two sides of each comparison are genuinely comparable; that the uncertainty budget of the prescribed procedure is given with its own statistical precision; that the leading attribution drift and the dominant uncertainties are explained, and tested by an intervention on the inputs, through a property of the input definitions; and that the coverage comparison is carried across every detector variation the release provides.

Throughout, the published ATLAS high-level network (hlDNN) is used only as an external reference point in the same fifteen-variable feature space. We do not attempt to reproduce it, and we

do not interpret the difference between our tagger and it as a measurement of anything.

2 Dataset and feature-space reconstruction

2.1 ATLAS Open Data Record 80030

The ATLAS top-tagging open dataset [13] is built from simulated proton–proton collisions at $\sqrt{s} = 13$ TeV passed through the full ATLAS detector simulation [23] in GEANT4 [24]. Signal jets come from a hypothetical $Z' \to t\bar{t}$ resonance with $m_{Z'} = 2$ TeV generated with PYTHIA 8 [25], reweighted at generation to populate the jet p_{T} range 0.35–5 TeV approximately uniformly; background jets come from QCD light-quark and gluon production [11]. Jets are clustered with the anti-k_{T} algorithm [26] at $R = 1.0$ using FastJet [27]. Each entry stores the four-vectors of up to 200 jet constituents, the jet four-vector, fifteen high-level substructure quantities, a signal/background label, and, for the nominal samples only, a training weight and a vector of PYTHIA shower weights.

The release distributes three families of samples. The *nominal* family supplies training and testing jets. Nine *detector-systematic* families apply a single detector variation to the constituents: cluster energy scale up and down (`esup`, `esdown`), cluster energy resolution (`cer`), cluster position resolution (`cpos`), track efficiency in two configurations (`teg`, `tej`), track fake rate in two configurations (`tfl`, `tfj`), and track momentum bias (`bias`) [11]. Six *modelling* families replace generator components: matrix-element and parton-shower matching for the signal (`ttbar_pythia` against `ttbar_herwig`), the Herwig 7 angular-ordered against dipole parton shower for the background (`angular` against `dipole`) [28], and the Sherpa 2.2 cluster against Lund-string hadronisation model for the background (`cluster` against `string`) [29].

Table 1 lists the seventeen samples processed here, with the file and row counts recorded by the read-only cache integrity audit. In total **80,481,421 jet entries** were read and reconstructed. The released `train_nominal` partition is far larger than the subset used here; we use 5,000,000 of its jets, a choice discussed in Sections 3.1 and 6.2. Each detector-systematic sample enters the evaluation with 5,000,000 jets, of which at most 244 fail the eligibility rule of Section 3.1; this is the size the dataset documentation recommends as a minimum in evaluation when the experimental uncertainties are calculated [30].

Table 1 The seventeen samples processed in this work, with the number of released HDF5 files and jet entries in each. Counts are taken from the read-only cache integrity audit, which verified file-by-file and row-by-row agreement between every source file and its reconstructed feature cache. The total is 80,481,421 jet entries in 810 cache files.

Family	Sample	Files	Jet entries
Nominal	`train_nominal`	50	5,000,000
	`test_nominal`	104	10,306,813
Detector	`esup`	50	5,000,000
	`esdown`	50	5,000,000
	`cer`	50	5,000,000
	`cpos`	50	5,000,000
	`teg`	50	5,000,000
	`tej`	50	5,000,000
	`tfl`	50	5,000,000
	`tfj`	50	5,000,000
	`bias`	50	5,000,000
Modelling	`ttbar_pythia`	2	193,792
	`ttbar_herwig`	2	180,811
	`cluster`	51	5,000,004
	`string`	51	5,000,001
	`angular`	50	4,900,000
	`dipole`	50	4,900,000
Total		**810**	**80,481,421**

2.2 The fifteen high-level quantities

We use the fifteen high-level quantities of Table 3 of Ref. [11]: N-subjettiness τ_1–τ_4 computed on winner-take-all k_{T} axes [3, 4, 31]; the k_{T} splitting scales $\sqrt{d_{12}}$ and $\sqrt{d_{23}}$ [32]; the energy correlation functions ECF_1, ECF_2, ECF_3 and the ratios C_2 [5] and D_2 [33, 34]; the generalised energy-correlation ratios L_2 and L_3 built from the observables of Ref. [35] in the combination specified by ATLAS [36]; the minimum pairwise subjet mass Q_w [37]; and the jet thrust major [38]. No kinematic variable such as jet p_{T} or mass enters the classifier, which keeps the feature space identical to the published high-level baseline.

2.3 Why the released high-level branches cannot be used

The fifteen high-level branches are present in the systematically varied files, and a weight- or

branch-based robustness study would naturally read them. In the four calorimeter-cluster samples for which exact jet pairing is valid, they do not carry the variation. Against `test_nominal`, `esup`, `esdown` and `cpos` each have 5,000,000 exact matches, and `cer` has 4,995,059. For every valid matched jet, all fifteen stored high-level values are float32 bit-identical to their nominal counterparts. The released constituents nevertheless differ: the relative summed-constituent-p_{T} shift under `esup` has mean +6.780753% and median +7.406764%; under `esdown` its mean is −6.780749% and median −7.406822%. The `cer` summed-energy response has an RMS of 3.770123%, while the `cpos` constituent-centroid displacement has median $\Delta R = 4.9387 \times 10^{-5}$. Thus the stored high-level branches of these four samples would conceal a real constituent-level detector variation.

The dataset documentation anticipates part of this. It states that in the samples meant for assessing experimental systematic uncertainties the jet four-vector "is not re-calculated from the systematic varied jet constituents", because the jet four-vector is calibrated [30]. It makes no corresponding statement about the fifteen high-level quantities, which it describes only as summaries of the constituent information. The audit shows that, in the four pairable calorimeter samples, they behave like the jet four-vector: they are identical to their nominal values and do not follow the varied constituents. We document this as a property of the release rather than as an error in it, but it means that the stored branches cannot be used to measure a tagger's response to these variations.

Because this observation is what makes the rest of the paper necessary, it is documented sample by sample in Supplementary Table S1. Exact-key hits in the four track samples and `bias` are diagnostic only: their full samples do not pass the validated one-to-one pairing criterion, so no pairwise high-level identity fraction or same-jet constituent shift is reported for them. Their constituent multiplicities are described as unpaired sample distributions; `bias` has 4,167,067 diagnostic exact-key hits but falls below the pairing floor (Section 4.4).

A second point of interpretation is worth stating explicitly, because it is easy to get wrong. The nominal files carry an `EventInfo_mcEventWeights` array. These are renormalisation- and factorisation-scale and PDF weights — *theory* weights. They are not detector systematics, and the detector systematics of this release are the dedicated samples listed in Table 1. In this work the shower-weight array is used only for the four initial- and final-state-radiation scale variations of Section 4.1, exactly as the reference implementation prescribes.

2.4 Reconstruction and closure

All fifteen quantities are recomputed from the released constituent four-vectors by a single NumPy/Numba implementation [39, 40] applied identically to every sample. Two implementation details determine whether the reconstruction closes. Every angular separation is measured in rapidity, not pseudorapidity, matching the convention of the FastJet interfaces [27] the ATLAS tools call; because the released constituents carry non-zero mass, the two differ. And thrust major is the ordinary three-dimensional thrust of the constituents boosted to the jet rest frame, evaluated with the same seed-and-search policy as the ATLAS reference implementation [30], not a transverse axis perpendicular to the thrust axis.

The reconstruction is validated against the stored nominal branches on 14,103 jets drawn from `test_nominal` and the six modelling samples, the seven samples in which the stored branches are meaningful. Per quantity, the median relative deviation lies between 2.5×10^{-8} (ECF_1) and 4.8×10^{-7} (Q_w). Of the $14{,}103 \times 15 = 211{,}545$ quantity–jet comparisons, 153 (0.072%) exceed a relative tolerance of 10^{-3}; thrust major contributes none of them and has a maximum relative deviation of 4.6×10^{-4}.

Those 153 residual comparisons fall on 44 physical jets, and an independent forensic pass classified every one of the 44 into one of four exhaustive classes. The counts are: **0** proven formula or implementation mismatches; **1** reproducible numerical or floating-point degeneracy; **43** jets consistent with information lost in the export; and **0** unresolved. It is the per-jet classification that carries these labels, not the 153 individual comparisons, and the distinction matters: it is not the case that all 153 tolerance failures are information-loss cases, but rather that 43 of the 44 affected jets are, with the remaining jet accounted for by a deterministic float32 rounding-cell control that reproduces its single Q_w residual.

For the 43 information-loss jets the argument is specific. The complete float32 interval of the stored ECF_1 lies above the maximum scalar p_T sum permitted by any rounding of the released constituents. Since ECF_1 is exactly $\sum_i p_\mathrm{T}^{(i)}$, no choice of arithmetic order, angular convention, clustering or formula can recover that missing momentum; these are jets for which the released constituents do not carry all the information the stored branch was computed from. Thirteen constituent orderings, a C++-order float64 replay, float32 and mixed-precision accumulation controls and an independent full-rescan exclusive-k_T replay each repaired zero of them. On that basis the reconstruction was frozen. It is identified by a fingerprint, `c28a2625...`, that covers the observable kernels and the input handling.

Six of the fifteen quantities carry an overall unit convention that differs from the stored branches by a fixed power of 10^6 ($\sqrt{d_{12}}$, $\sqrt{d_{23}}$, ECF_1, Q_w; 10^{12} for ECF_2; 10^{18} for ECF_3), because the reconstruction works in MeV throughout. The factor is a fixed monotone rescaling of a single input, applied identically to every sample, so it does not affect a rank-splitting tree model, its TreeSHAP attributions, or any nominal-to-varied comparison. It is recorded here because the released feature cache uses the MeV convention.

2.5 Frozen cache and integrity audit

Reconstruction produced 810 cache files. A read-only audit compared every cache file to its source: file counts, per-file row counts, presence of all fifteen quantities, and the reconstruction fingerprint. Every file matched, with zero mismatches and a single fingerprint across all 810 files.

The audit identified 284 non-finite values in the entire 80.5-million-entry campaign, all in L_3 and all at the same deterministic low-multiplicity boundary, where the generalised three-point correlator vanishes and L_3 becomes $-\infty$. The `cer` sample additionally contains 4,745 entries that ATLAS itself flagged in the stored L_3 branch with a -999 sentinel; 198 of these coincide with the L_3 boundary. No cache value was altered to suppress either case. Jets are removed from training and evaluation only by the fixed eligibility rule of Section 3.1, and the removed counts are reported per sample.

3 Tagger training and evaluation

3.1 Training

The classifier is gradient-boosted decision trees, XGBoost [14] version 3.2.0, with the binary logistic objective (`binary:logistic`) and the histogram tree method (`tree_method="hist"`). The non-default hyperparameters are 500 trees, maximum depth 6, learning rate 0.1, row subsample fraction 0.8 and column subsample fraction per tree 0.8; `eval_metric` is set to `logloss` and the random seed to 42. Every remaining hyperparameter is left at the library default for that version, including minimum child weight 1, $\gamma = 0$, L_2 regularisation $\lambda = 1$, L_1 regularisation $\alpha = 0$, maximum delta step 0, and column subsampling per level and per node of 1. The full software environment is Python 3.11.16, NumPy 2.4.6, Numba 0.67.0, scikit-learn 1.9.0, h5py 3.16.0, SciPy 1.17.1 [41], joblib 1.6.0 and SHAP 0.51.0. Gradient boosting splits on rank, so it is invariant to the monotone rescalings the feature space needs — ECF_3 spans many orders of magnitude — and requires no standardisation.

Training uses 5,000,000 `train_nominal` jets from the first fifty released files of that partition, read exclusively from the frozen feature cache. Eligibility is fixed and applied without imputation: a jet enters if all fifteen reconstructed quantities are finite and it has at least three constituents, the minimum multiplicity already required by the ATLAS jet selection [11]. Three of the five million jets fail this rule, leaving 4,999,997. The dataset's stated rule that every jet's contribution to the loss be weighted by its training weight is honoured: the released `training_weights` are passed as the per-jet `sample_weight` argument of the fit, and are used nowhere else — every performance number reported below is unweighted.

A held-out validation subset is assigned by a deterministic SplitMix64 [42] hash of the Monte Carlo event number at a nominal validation fraction of 0.1, so that every jet sharing an event number falls on the same side and no event is split across the boundary. The result is a five-million-jet nominal subset split into 4,499,612 jets used for fitting and 500,385 held out for validation and, later, for conformal calibration; the realised validation fractions are 10.024% for background and 9.992% for signal.

Three properties of this procedure are worth stating explicitly, because they bound what the held-out numbers mean. First, the hyperparameters above were fixed before fitting and no search over them was performed: the only other tagger trained in the frozen campaign is a 4,000,000-jet pilot with an identical configuration, retained unchanged, and the move to five million jets was a choice of training-set size, not a selection against a validation score. Second, the fit is called without an evaluation set, so no early stopping, no boosting-round selection and no threshold tuning takes place; the number of trees is the fixed 500 above. Third, in consequence, the 500,385 validation jets influenced neither hyperparameter selection nor the choice of final model, and they are therefore independent of the fitted classifier in the sense the conformal calibration of Section 4.5 requires.

The training-set size is guided by the reference workflow of the dataset rather than by a performance optimisation. The documentation's example training script uses six million of the training jets and advises taking as many as fit in memory, while stating that state-of-the-art performance will require training on the entire partition [30]; its five-million-jet recommendation refers to evaluation, which Section 2.1 follows. Our five-million-jet training subset is therefore close to the six million the reference script uses and well short of the full partition, and Section 6.2 discusses what that implies for the comparison with the published reference.

The unweighted validation area under the receiver-operating-characteristic curve (AUC) is 0.93190, against 0.93199 on the independent test sample. These held-out validation and test values show no evidence of an obvious performance discrepancy; we do not read a stronger statement than that into them, since a single pair of AUC values cannot exclude the milder forms of overfitting.

3.2 Evaluation protocol

Inference is run over the nominal test sample, the nine detector-systematic samples and the six modelling samples, always through the same loader, the same fixed feature order, and the same eligibility rule. On `test_nominal`, 10,306,812 of 10,306,813 entries are eligible; across the detector samples the number removed ranges from 1 to 244 (`cer`), and across the modelling samples from 0 to 5. Row alignment between each prediction file and its source cache was verified chunk by chunk after writing.

Two properties of the samples matter for interpretation. Training weights are used *only* during fitting; every performance metric below is unweighted, with a vector of ones, for the nominal and detector-systematic samples, and shower-weight ratios are applied only to the theory variations. And the class composition of the varied samples is not uniform: the calorimeter-cluster samples and `bias` sit near 51.8–52.0% signal, close to the nominal 50.5%, whereas `teg` (30.0%), `tej` (31.8%), `tfj` (32.0%) and `tfl` (18.5%) do not. The track samples are therefore different jet populations, not re-simulations of the same jets, and comparisons involving them are made at the level of whole datasets rather than matched jets. This is also why any quantity that averages over classes — overall conformal coverage in particular — has to be read against the class mixture of the sample it was measured on.

4 Methods

4.1 Systematic uncertainties

Performance and its uncertainty follow the procedure the record prescribes, transcribed from the reference implementation at the frozen commit `5bc3c8d7` [30]. The two reference scripts at that commit, `calc.py` and `plot_everything.py`, are preserved alongside the analysis code with their git blob and SHA-256 hashes recorded, so the transcription can be checked line by line rather than taken on trust. Metrics are computed with scikit-learn [43]: unweighted AUC, accuracy at a score threshold of 0.5, and inverse background efficiency $1/\epsilon_{\mathrm{bkg}}$ at fixed signal efficiency. The primary working point for uncertainties is 50% signal efficiency; 80% is additionally quoted for comparison with the published benchmark, and no systematic uncertainty is evaluated at the 80% point — every uncertainty in this paper refers to 50% signal efficiency. Background rejection is evaluated inclusively and in fourteen p_{T} bins of 200 GeV spanning 350–3150 GeV.

The uncertainty from a detector variation is

$$u_s = \left| \frac{1/\epsilon_{\mathrm{bkg}}^{s}}{1/\epsilon_{\mathrm{bkg}}^{\mathrm{nom}}} - 1 \right|, \tag{1}$$

evaluated inclusively and in each p_{T} bin. Equation 1 is a dataset-level comparison: no jet pairing is required or used, which is why samples that do not pair jet-by-jet are still usable here. Two-sided variations are combined into an envelope, $u = \max(u_{\uparrow}, u_{\downarrow})$, and independent components are added in quadrature. Cluster energy scale is the `esup`/`esdown` envelope; the cluster total adds `cer` and `cpos` in quadrature. Track efficiency is the `teg`/`tej` envelope, track fake rate the `tfl`/`tfj` envelope, and the track total adds `bias` in quadrature.

Modelling uncertainties are two-point comparisons between alternatives rather than comparisons to nominal: signal modelling from `ttbar_herwig` against `ttbar_pythia`, background parton shower from `dipole` against `angular`, and background hadronisation from `cluster` against `string`. Because the single-class modelling samples cannot form a receiver-operating-characteristic curve on their own, the nominal jets of the missing class are appended before evaluation, as the reference implementation does. The Standard-Model $t\bar{t}$ samples run out of statistics above 1.5 TeV, so signal modelling is measured in the bins below that edge and held at its maximum measured value above. Two properties of this term should be read together with its size: it is measured on Standard-Model $t\bar{t}$ and applied to a signal sample that is a 2 TeV Z' resonance, and above 1.5 TeV it is an extrapolation rather than a measurement. Both follow the procedure the record prescribes, and we adopt it unchanged, but the dominant entry in the uncertainty budget is therefore a transfer between processes. We follow the official plotting convention exactly, including its treatment of the 1.35–1.55 TeV interval, which straddles the 1.5 TeV edge and receives the closing value of the measured range rather than the extrapolated maximum; this is stated here because it is visible in Fig. 2 and in the supplementary bin-by-bin tables.

The four initial- and final-state-radiation (ISR and FSR) scale variations are the only use of the shower-weight array. They are applied as weight ratios to the nominal predictions, separately for signal and for background, and combined as up/down envelopes and then in quadrature.

4.2 Statistical precision and training-seed dependence

The dataset documentation states the precision requirement explicitly: a systematic difference is meaningful only if the statistical uncertainty on the performance metric is smaller than the difference itself [30]. We measure that statistical uncertainty with a Poisson bootstrap [44, 45]. Every jet of every evaluated sample receives a weight drawn from a Poisson distribution of unit mean, and the complete chain of Section 4.1 — working-point thresholds, background rejections, envelopes, quadrature sums, the modelling complements and the extrapolation of signal modelling — is recomputed with those weights. The correlations on which the uncertainty definition relies are preserved. A jet of `esup`, `esdown`, `cer` or `cpos` that matches a `test_nominal` jet on the exact key of Section 4.4 takes the weight of its nominal partner, so that the statistical fluctuation of the jets common to both samples cancels in the comparison. The nominal reference uses all 10,306,812 test jets, of which only about five million have a partner, so the remaining nominal jets fluctuate independently and set a statistical floor below which a component cannot be resolved with these sample sizes. The four ISR/FSR variations, which are weight ratios on the nominal predictions, and the nominal jets appended to the single-class modelling samples reuse the nominal weights. The track samples, `bias`, the modelling samples and the few unmatched `cer` jets are resampled independently. With unit weights the procedure reproduces every frozen number exactly, and this is checked before any replica is drawn. We use 100 replicas with a fixed seed and quote the standard deviation of each quantity over them as its statistical uncertainty.

Each uncertainty component is a modulus, $|r - 1|$, and is then combined in envelopes and quadrature, so it is positive by construction and statistical noise raises its expected value. For a component whose true size is small, the bootstrap distribution is skewed and its mean lies above the measured value; the measured value itself carries the same upward bias. We therefore keep the frozen values as the central values, use the bootstrap only for their spread, and state the replica mean where this bias matters.

Training-seed dependence is measured by refitting the frozen configuration with five further seeds

on exactly the same 4,499,612 fitting jets; the seed controls the row and column subsampling of the boosting, and the validation split is unchanged. The refit with seed 42 is checked to reproduce the frozen model byte for byte, with the same SHA-256 and bit-identical predictions on every sample. Each refit is evaluated with the frozen chain on the same test and systematic samples. These fits, and the input-set variants of Section 4.3, were made after the analysis was frozen; none of them replaces the frozen model, and no frozen number is changed by them.

4.3 Momentum-scale degree of the inputs

Under a common rescaling $p^{(i)} \to \lambda\, p^{(i)}$ of every constituent four-momentum, each of the fifteen quantities scales as λ^d with a fixed integer degree d that follows from its definition. The four N-subjettiness quantities, which are normalised by the summed constituent p_{T}, and C_2, D_2, L_2, L_3 and thrust major are dimensionless ($d = 0$); $\sqrt{d_{12}}$, $\sqrt{d_{23}}$, $\mathrm{ECF}_1 = \sum_i p_{\mathrm{T}}^{(i)}$ and Q_w have $d = 1$; ECF_2 has $d = 2$ and ECF_3 has $d = 3$. A variation that shifts the constituent energy scale by a small fraction k therefore moves a quantity of degree d by about dk, and leaves the dimensionless quantities nearly unchanged.

Two measurements test this. First, for every exactly paired calorimeter jet we compute the per-jet response $X^s/X^{\mathrm{nom}} - 1$ of each reconstructed input, on all matched pairs rather than on the SHAP sample. A relative change is the natural measure for the degree argument, but it is not a measure of the effect on the classifier: a tree model splits on rank, so what matters to it is the shift of an input relative to that input's spread and to the split points the model uses. Second, the frozen configuration is retrained, with the same six seeds and on the same fitting jets, on two reduced input sets: the nine dimensionless quantities alone, and those nine together with the five dimensionful quantities other than ECF_1, each divided by ECF_1^d, which removes their momentum scale but keeps their shape. Both are evaluated with the frozen chain. These variants are diagnostics of where the sensitivity of the frozen tagger comes from; they are not proposed as alternative taggers.

4.4 Paired SHAP-Delta

For a jet present in both the nominal and a varied sample, the SHAP-Delta of feature i is

$$\Delta\mathrm{SHAP}_i \;=\; \phi_i(\mathbf{x}^{\,s}) - \phi_i(\mathbf{x}^{\,\mathrm{nom}}), \qquad (2)$$

with ϕ_i the exact TreeSHAP attribution [17] of the frozen model, evaluated in the additive margin space of the booster. Equation 2 is defined per jet, so it exists only for samples that pair. Two summaries are reported and are deliberately *not* interchangeable: the mean of $|\Delta\mathrm{SHAP}_i|$ over jets, which measures how much a feature's attribution moves for individual jets, and the relative change in the global mean $|\phi_i|$, which measures how much the feature's average importance moves. A feature can have a large per-jet drift with almost no change in average importance, and the two are tabulated separately. Classifier-score drift is reported separately again, in probability space, and is computed under the same class mask as the SHAP summary it accompanies.

The comparison requires that the two sides be the same physical jet. Jets are matched on the exact key (event number, jet η, jet ϕ, label), with η and ϕ compared as float32 bit patterns rather than with a tolerance, and with negative zero folded onto positive zero. Uniqueness of the key and one-to-one matching are re-checked on every run. The pairing index of the campaign that produced the frozen paired frames was built over both nominal partitions, and the frozen logs record that every matched jet came from `test_nominal` and none from `train_nominal`: no jet the model was fitted on entered a paired comparison. The pairing code now indexes `test_nominal` alone, so that guarantee is enforced rather than observed.

Table 2 gives the validated pairing outcome and diagnostic key-hit coverage on the full five-million-jet detector samples, not on a shard. Four pair essentially completely against `test_nominal`: `esup`, `esdown` and `cpos` at 100.000%, and `cer` at 99.901%, where 96.0% of the 4,941 unmatched jets carry the -999 sentinel ATLAS wrote into the stored L_3 branch for exactly those jets. The paired SHAP-Delta analysis is restricted to these four, and exact pairing is claimed only for them.

The remaining five detector samples are excluded on evidence, and the full-sample stored-branch audit sharpens the reason. `teg` carries

Table 2 Validated pairing and diagnostic exact-key hits against `test_nominal` on (event number, jet η, jet ϕ, label). Rates are for the whole five-million-jet sample. Only the four calorimeter samples are pairable. The `bias` rate is diagnostic, below the pairing-coverage floor. The four track samples were not validated for pairing in the paired-analysis campaign: their signal fractions are far from the nominal 50.52%, so they represent different jet populations. The later stored-branch audit records diagnostic exact-key hits for them but does not treat those hits as valid pairs or compute pairwise identity fractions. The signal column is the fraction of the whole eligible sample, not of the matched subset. The four calorimeter rates come from the paired campaign (pair fingerprint `c91534fd`); the `bias` rate comes from an earlier run (`7be1a72a`) whose paired output was subsequently archived as obsolete, and the track hits come from the full-sample stored-branch audit.

Sample	Signal [%]	Exact-key hits	Hit [%]
`esup`	51.85	5,000,000	100.000
`esdown`	51.85	5,000,000	100.000
`cpos`	51.85	5,000,000	100.000
`cer`	51.81	4,995,059	99.901
`bias`	51.96	4,167,067	83.341
`teg`	30.01	2,250,423	45.008
`tej`	31.80	1,906,619	38.132
`tfj`	31.98	2,006,933	40.139
`tfl`	18.46	2,749,571	54.991

183,802 duplicate systematic keys, so an exact one-to-one join is ambiguous for it whatever its coverage; `tej`, `tfj` and `tfl` reach exact-key coverages of 38.1%, 40.1% and 55.0%, all far below this dataset's validated floor. The four track samples also have the class compositions of Table 2 and are plainly different jet populations. `bias` is not pairable on the full sample: 4,167,067 of 5,000,000 jets, or 83.341%, register an exact-key hit, far below this dataset's 99.95% floor. The key-hit subset is 62.34% signal against 51.96% for the sample as a whole, so jets without a key hit are strongly background-enriched. A forensic diagnostic on one 100,000-jet shard, run against the identity columns of all 104 nominal files, decomposes that shortfall and reproduces the full-sample rate closely at 83.223%: 12.095% of the shard's jets carry event numbers absent from `test_nominal` altogether, and of the 4.682% that share an event number without matching, those with a same-label nominal candidate have a median nearest-neighbour separation of $\Delta R = 1.38$ and no candidate agreeing in either the η or the ϕ float32 bits, so their axes differ physically rather than by rounding. `bias` therefore enters the dataset-level uncertainty of Section 4.1 and the dataset-level coverage of Section 4.5, where pairing is not required, and is excluded from the paired analysis.

TreeSHAP is evaluated on a sample of 100,000 pairs per systematic, drawn without replacement and stratified jointly on class and p_{T} bin, with nested subsets of 25,000 and 50,000 retained for a convergence check.

4.5 Conformal coverage

Split conformal prediction [18, 19] converts a score into a prediction *set* with a calibrated coverage level. We use the class-conditional form. The nonconformity score of a jet with respect to class c is $s_c = 1 - \hat{p}_c$, and a threshold q_c is fixed for each class as the $\lceil (n_c + 1)(1 - \alpha) \rceil$-th order statistic of the calibration nonconformities of jets whose true class is c, at $\alpha = 0.1$. Class c is included in the prediction set of a jet when that jet's nonconformity with respect to c does not exceed the threshold,

$$1 - \hat{p}_c \;\leq\; q_c \qquad \Longleftrightarrow \qquad \hat{p}_c \;\geq\; 1 - q_c, \quad (3)$$

so a prediction set can contain one class or both. The inequality runs this way round because s_c is a nonconformity, which is small for a confident, correct prediction: the set collects the classes the jet is *not* unusual for.

Calibration is performed *once*, on the frozen held-out nominal validation split of Section 3.1 — 500,385 jets, 252,215 signal and 248,170 background — giving $q_{\mathrm{sig}} = 0.5700$ and $q_{\mathrm{bkg}} = 0.6447$. By Eq. 3 the signal class is included when $\hat{p}_{\mathrm{sig}} \geq 0.4300$ and the background class when $\hat{p}_{\mathrm{bkg}} \geq 0.3553$, that is when $\hat{p}_{\mathrm{sig}} \leq 0.6447$. The two acceptance regions overlap, which is why the empty prediction set is not merely rare but impossible here, and why the observed empty-set rate is exactly zero on every sample. As Section 3.1 records, these calibration jets took no part in fitting the model and no part in selecting its hyperparameters or in choosing between models, so the calibration set is independent of the classifier it calibrates.

Those thresholds are then applied unchanged to the nominal test sample and to each of the nine detector-systematic samples. Coverage is measured empirically and unweighted; no refitting,

recalibration or reweighting is performed at any point.

Two points of interpretation are needed. First, split conformal guarantees marginal coverage only when calibration and evaluation data are exchangeable, which a systematic variation breaks by construction [46]. The quantity measured here is therefore not a guarantee but a diagnostic: how far the empirical coverage of a nominally calibrated tagger moves when the detector response is varied. Second, because the calibration is class-conditional, the primary quantities are the two class-conditional coverages, and those are the numbers to compare across samples. The overall coverage is a weighted average of the two with weights given by the class mixture of the sample, and the mixtures here range from 50.5% signal on nominal to 18.5% on `tfl`. Overall coverage is therefore reported as a secondary, descriptive number, and a change in it between two samples cannot be attributed to the detector variation without first accounting for the change in mixture.

5 Results

5.1 Nominal performance

Table 3 gives the frozen nominal result on the 10,306,812 eligible test jets. The tagger reaches an unweighted AUC of 0.9320, accuracy 0.8559, and background rejection 41.91 at 50% signal efficiency and 9.29 at 80%. Their statistical uncertainties, from the bootstrap of Section 4.2, are ±0.00008 on the AUC, ±0.12 on the background rejection at 50% and ±0.015 at 80%. Across the six training seeds the background rejection at 50% ranges from 41.74 to 41.91, with a standard deviation of 0.07, so the seed dependence of the nominal performance is smaller than its statistical uncertainty. The published hlDNN values, taken from Table 4 of Ref. [11], are listed beside them purely as an external reference in the same fifteen-variable feature space. We did not train that network, did not attempt to reproduce it, and the difference between the two rows is not a measurement: the architectures differ, and our training set is a small fraction of the released nominal partition. Fig. 1 shows the full receiver-operating-characteristic curve. The two working points quoted above are marked on it; between them the curve is smooth, so neither is a special feature of the classifier. The curves for the two cluster energy-scale variations lie just below the nominal one over the whole range — their areas under the curve are 0.9294 (`esup`) and 0.9286 (`esdown`), against 0.9320 nominal — which is the same ordering that the background-rejection uncertainties of Section 5.2 report at a single working point. Panel (a) of Fig. 2 shows the p_{T} dependence of background rejection, which falls from 93.4 in the 350–550 GeV bin to 28.6 in the 2950–3150 GeV bin.

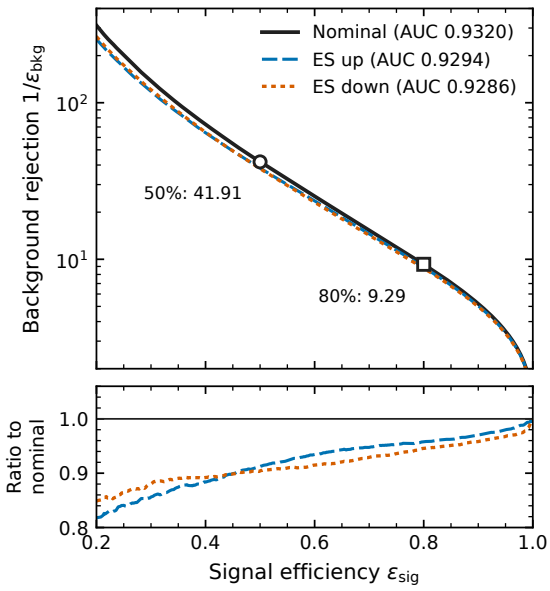


Fig. 1 Receiver-operating-characteristic curve of the frozen tagger on the nominal test sample, and on the two cluster energy-scale variations, drawn as background rejection $1/\epsilon_{\mathrm{bkg}}$ against signal efficiency ϵ_{sig} on a logarithmic vertical axis. The horizontal range is restricted to $\epsilon_{\mathrm{sig}} > 0.2$, below which the rejection rises steeply and carries no information relevant to the working points used here. The lower panel gives the ratio of each varied curve to the nominal one on a linear axis; on the logarithmic axis above, the three curves are too close to separate by eye, and the ratio introduces no quantity that is not already in the upper panel. Curves are computed from the frozen prediction files with the unweighted convention of Section 4.1; the areas under the curve quoted in the legend are the frozen values, the nominal one as in Table 3 and the two varied ones as quoted in Section 5.1. Open markers are the two working points reported in this paper, labelled with the frozen background-rejection values of Table 3 rather than with values read off the curve.

5.2 Systematic uncertainties

Table 4 gives the inclusive breakdown at 50% signal efficiency and Fig. 2(b) its p_{T} dependence. The

Table 3 Nominal XGBoost performance. The ATLAS hlDNN row is a published external reference in the same 15-variable high-level feature space; it is not a reproduction. The two indented rows give the statistical standard deviation of each metric over 100 Poisson bootstrap replicas of the test sample, and the standard deviation over six training seeds of the same configuration on the same training jets (Section 4.2); neither is a confidence interval on the published reference.

Model	Role	AUC	Accuracy	BR@50%	BR@80%
XGBoost (this work)	Frozen final result	0.931995	0.855854	41.9064	9.28847
statistical s.d.	Bootstrap, 100 replicas	±0.000080	±0.00010	±0.12	±0.015
training-seed s.d.	Six seeds, same training jets	±0.000023	±0.000041	±0.073	±0.0061
ATLAS hlDNN	Published external reference; not reproduced	0.9374 ± 0.0001	0.8628	47.2 ± 0.4	10.36 ± 0.03

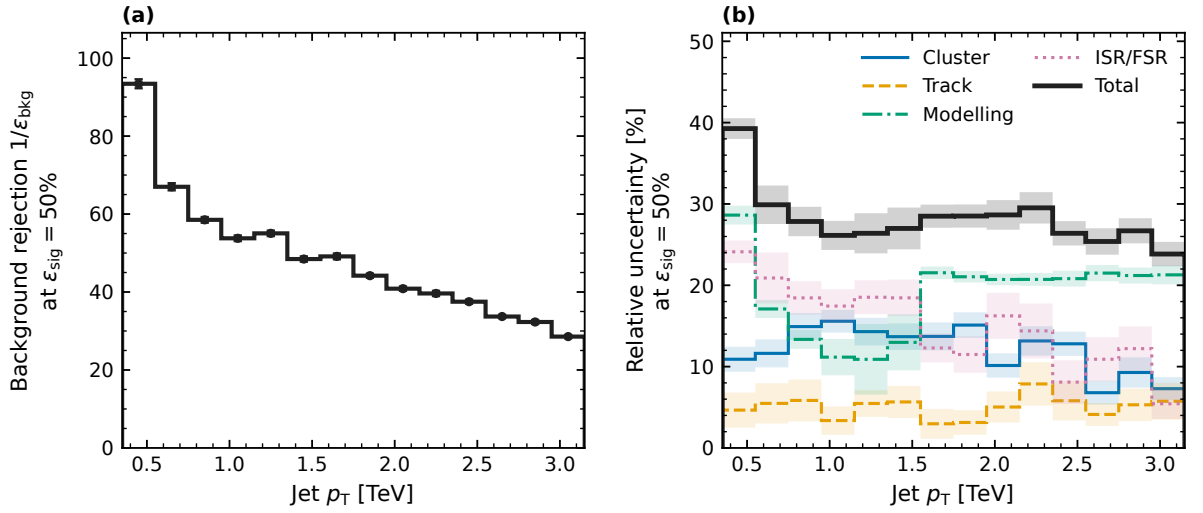


Fig. 2 Tagger performance and its systematic uncertainty as a function of jet p_T, at 50% signal efficiency throughout, in the fourteen physical 200 GeV bins spanning 350–3150 GeV. **(a)** Nominal background rejection $1/\epsilon_{bkg}$; markers are bin centres. **(b)** Relative uncertainty on background rejection by category: cluster (energy scale envelope combined in quadrature with energy and position resolution), track (efficiency and fake-rate envelopes combined in quadrature with track bias), modelling (signal and background parton shower and hadronisation), and ISR/FSR (the four shower-weight scale envelopes); Total is their quadrature sum. The error bars in (a) and the shaded bands in (b) are ±1 standard deviation of 100 Poisson bootstrap replicas, drawn around the frozen measured values (Section 4.2); their bin-by-bin values are given in Supplementary Table S8. The step in the modelling curve between the 1.35–1.55 TeV and 1.55–1.75 TeV bins is the official extrapolation convention described in Section 4.1: signal modelling is measured only below 1.5 TeV and is held at its maximum measured value from 1.55 TeV upwards. Bin-by-bin values for every component are given in Supplementary Tables S2 and S3.

total relative uncertainty on background rejection is 25.83%.

Generator modelling dominates at 20.01%, and within it the signal generator comparison (`ttbar_herwig` against `ttbar_pythia`) contributes 18.59% against 7.40% for the two background comparisons combined. The ISR/FSR scale variations follow at 12.04%, almost entirely from the signal final-state-radiation envelope (11.34%); the three remaining scale envelopes — signal ISR 0.19%, background ISR 1.87% and background FSR 3.60% — combine to 4.06% in quadrature, which is not negligible beside the detector terms even though it is small beside the signal FSR envelope. Detector effects are smaller: 10.07% for the calorimeter cluster category, of which the energy-scale envelope alone is 9.58%, and 4.51% for tracking, split between the fake-rate (3.78%) and efficiency (2.47%) envelopes with a negligible 0.16% from track bias.

The p_T dependence in Fig. 2(b) is not flat. The total is largest in the lowest bin (39.3%), falls to about 26% between 0.95 and 1.55 TeV, and rises again above 1.55 TeV as the extrapolated signal-modelling term takes over. The cluster and track curves fluctuate bin to bin at the few-percent level.

The bootstrap of Section 4.2 separates that structure from sample statistics. In each bin the statistical standard deviation of the total is 1.3 to

2.6 percentage points (Supplementary Table S8). A χ^2 comparison of each binned detector curve with a p_{T}-independent constant, treating the fourteen bins as independent and the bootstrap standard deviations as Gaussian, gives $\chi^2 = 55.7$ for 13 degrees of freedom for the energy-scale envelope and 45.7 for the cluster category, so their p_{T} dependence is genuine. For the track category it gives $\chi^2 = 5.4$, and for `cer`, `cpos` and track bias 15.9, 3.4 and 4.7: the bin-to-bin fluctuation of these curves is compatible with statistics alone, and in eight of the fourteen bins the `cpos` and the track-bias values are each smaller than their own statistical standard deviation. The values of χ^2 for these small components are lower than 13 degrees of freedom would make likely, which is itself a sign that taking the modulus compresses their bin-to-bin scatter; for them the test is qualitative. Because every component is a modulus, noise also biases it upward: the replica mean of the binned total lies 0.8 to 1.6 percentage points above the measured value.

Inclusively, the statistical precision is sufficient for nearly every component (Table 4). The energy-scale envelope is $9.58 \pm 0.33\%$, signal modelling $18.59 \pm 0.50\%$, signal FSR $11.34 \pm 0.64\%$ and the total $25.83 \pm 0.58\%$, and every entry of the table except two lies at least five standard deviations from zero. Of the two background modelling comparisons that the table combines, the parton-shower one is $1.29 \pm 0.40\%$. The exceptions are `cpos`, at $1.13 \pm 0.33\%$, and track bias, at $0.16 \pm 0.32\%$ with a replica mean of 0.45%; track bias is consistent with zero and does not meet the documentation's requirement that the statistical uncertainty be smaller than the difference it measures [30].

The dependence on the training seed is of the same order as the statistical precision of the total and smaller than it for the nominal performance. Over six seeds the total is $26.32 \pm 0.47\%$, ranging from 25.83% to 27.01%, and the modelling term $20.65 \pm 0.43\%$; the frozen seed gives the lowest total and the lowest modelling term of the six, about one standard deviation below the six-seed means. The two effects are not independent, since a refitted model places a different set of jets near the working-point threshold, and we do not combine them. Together they indicate that the frozen total is determined to about 0.6 percentage points statistically and a further 0.5 from the choice of seed, and none of the conclusions drawn from the budget depends on either. The per-seed values are listed in Supplementary Table S9.

5.3 Paired SHAP-Delta

Fig. 3 and Table 5 give the paired attribution result on 100,000 matched pairs per systematic, for the four calorimeter-cluster variations of Table 2.

The per-jet attribution drift is concentrated. For the two energy-scale variations the largest mean $|\Delta\mathrm{SHAP}|$ falls on ECF_3 (0.127 for `esup`, 0.132 for `esdown`, in margin units), more than twice the next-largest entry, followed by ECF_1, L_2, thrust major and τ_4^{WTA}. `cer` is led by ECF_3 at about half that size (0.062), with τ_4^{WTA} close behind. `cpos` differs in both size and shape: its largest drift is an order of magnitude smaller (τ_4^{WTA}, 0.013) and the energy correlators no longer lead. That the two variations which change constituent energies move ECF_3 and ECF_1 most, while the variation that changes constituent positions does not, is what one expects from what each variation does to the inputs, and Section 5.4 measures that input response directly.

Global feature-importance orderings are highly stable under all four variations, and change only minimally. On the 100,000-pair samples the Spearman correlation between the nominal and varied orderings is at least 0.986 across all twelve systematic-by-class combinations, and is exactly 1 in eight of them; restricted to the four all-jet orderings the minimum is 0.993. The two highest-ranked features are identical in every case. Movement inside the top four is confined to two adjacent swaps of ranks three and four, both under `esdown` — L_2 with thrust major for all jets, and L_2 with $\sqrt{d_{23}}$ for background jets — and the largest rank change anywhere is two positions. This is the point of separating the two summaries: attributions move appreciably for individual jets while the average importance ordering the model relies on barely moves. Per-feature values of both summaries are given in Supplementary Table S6, and the score- and rank-level summaries, which are properties of a systematic and class subset rather than of a feature, in Supplementary Table S7.

Score drift, in probability space, is small and class-dependent. Under `esup` the mean signed score change is -0.0029 for signal and $+0.0055$ for background — the two classes move towards each other, the direction consistent with the loss of background

Table 4 Inclusive relative systematic uncertainties at 50% signal efficiency. Envelopes are taken before quadrature combination. "Frozen" is the frozen result. "Stat. s.d." is the standard deviation of the same quantity over 100 Poisson bootstrap replicas in which exactly paired calorimeter jets share their nominal partner's weight. "Six seeds" is the mean and standard deviation over six training seeds of the frozen configuration on the same training jets; seed 42 reproduces the frozen column exactly (Section 4.2).

Category	Component	Frozen [%]	Stat. s.d. [%]	Six seeds [%]
Cluster	Cluster energy scale	9.58	0.33	9.45 ± 0.11
Cluster	CER	2.89	0.36	2.89 ± 0.10
Cluster	CPOS	1.13	0.33	0.80 ± 0.19
Cluster	Cluster total	10.07	0.43	9.91 ± 0.11
Track	Track efficiency	2.47	0.45	2.26 ± 0.15
Track	Fake rate	3.78	0.45	3.69 ± 0.10
Track	Bias	0.16	0.32	0.11 ± 0.08
Track	Track total	4.51	0.55	4.33 ± 0.12
Modelling	Signal modelling	18.59	0.50	19.15 ± 0.43
Modelling	Background modelling	7.40	0.37	7.72 ± 0.29
Modelling	Modelling total	20.01	0.49	20.65 ± 0.43
ISR/FSR	Signal ISR	0.19	0.03	0.20 ± 0.01
ISR/FSR	Signal FSR	11.34	0.64	11.43 ± 0.33
ISR/FSR	Background ISR	1.87	0.25	1.90 ± 0.10
ISR/FSR	Background FSR	3.60	0.51	3.85 ± 0.28
ISR/FSR	ISR/FSR total	12.04	0.65	12.22 ± 0.37
Total	Total uncertainty	25.83	0.58	26.32 ± 0.47

rejection under `esup` (8.8% inclusively; the two-sided energy-scale envelope reported in Table 4 is the larger `esdown` value, 9.58%). Under `esdown` both classes move down, signal further (-0.0113) than background (-0.0045). `cer` and `cpos` produce shifts an order of magnitude smaller.

The nested 25,000 and 50,000 subsets reproduce the 100,000-jet result closely: Spearman correlations of the per-feature mean $|\Delta\text{SHAP}|$ against the full sample are ≥ 0.989 in every case, and the five leading drift features agree in 23 of 24 comparisons. Because the smaller sets are nested inside the largest, this is a convergence check on sample size, not an independent replication, and we present it as such.

5.4 Momentum-scale degree and the scale-free test

Table 6 gives the per-jet response of every input on all matched pairs, and Supplementary Table S10 its full distribution. Under `esup` the summed constituent transverse momentum ECF_1 rises by a median 7.41%, and the response of every other input is ordered by its degree: +4.29% to +5.48% for the other three quantities with $d = 1$, +12.57% for ECF_2, +17.63% for ECF_3, and between -1.87% and +0.28% for the nine dimensionless ones. `esdown` mirrors this with opposite sign. The dimensionful responses are smaller than a uniform rescaling would give (15.36% for ECF_2 and 23.91% for ECF_3), and the dimensionless ones are not exactly zero. Both show that the variation is not a uniform rescaling of all constituents, which is consistent with the documentation's description of it as acting on the calorimeter-reconstructed constituents [30]; we have not checked which constituents are left unscaled. Under `cer` the median responses vanish but individual jets move, and ECF_3 again has the largest median absolute response of all fifteen inputs, 4.88%, against 3.59% for ECF_2 and at most 2.46% for any dimensionless input. Under `cpos` no input moves by more than 0.12% in median absolute value, and ECF_3 and τ_4^{WTA} respond almost equally (0.117% and 0.115%).

This accounts for the leading entry of the attribution pattern of Section 5.3. ECF_3, the input of highest degree, is the input whose value moves most for the same physical jet under the two energy-scale variations and under `cer`, and it is the input whose attribution moves most under all three. It does not account for the rest of the ordering. Under

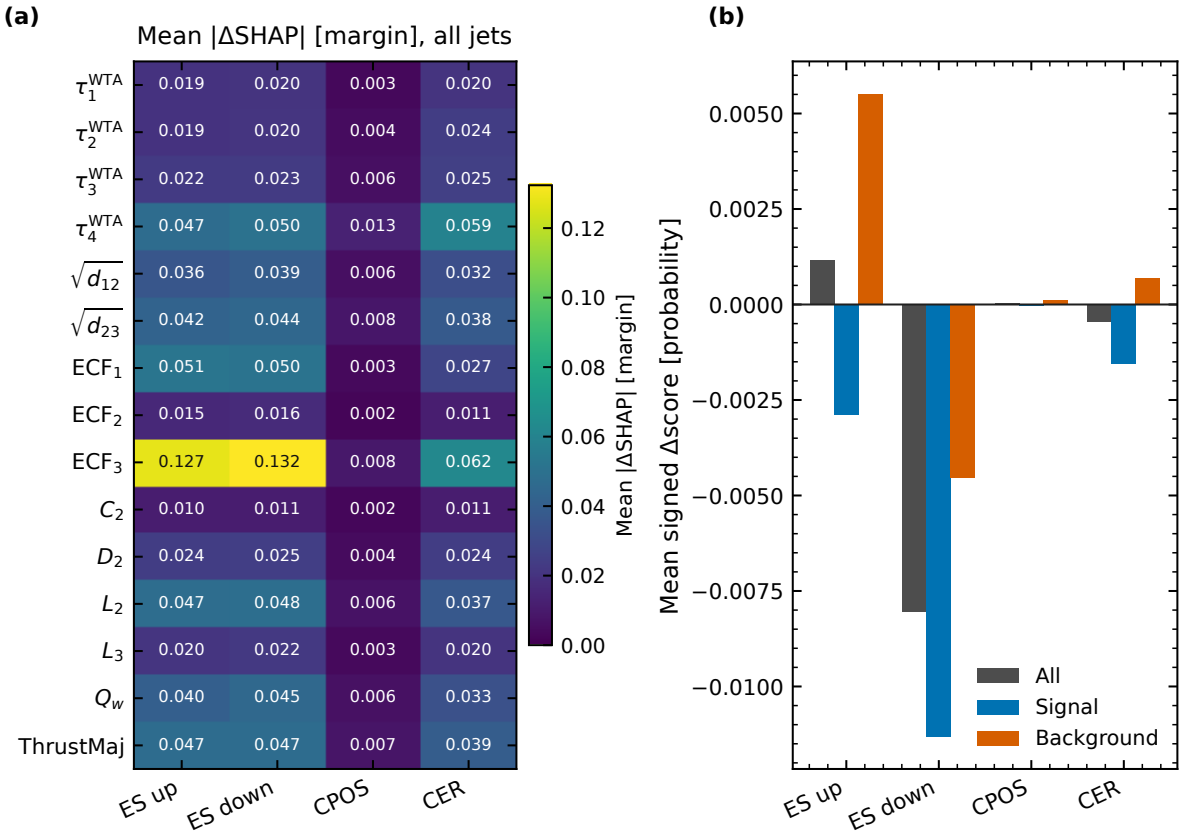


Fig. 3 Paired SHAP-Delta for the four calorimeter-cluster variations, on 100,000 matched nominal-to-varied jet pairs per systematic. Columns are, in order, `esup` (ES up), `esdown` (ES down), `cpos` (cluster position resolution) and `cer` (cluster energy resolution). **(a)** Mean |ΔSHAP| per feature over all jets, in the additive margin space of the booster; each cell carries its own value so that the small entries can be read without the colour scale. **(b)** Mean signed change in classifier score, in probability space, for all jets and separately for the signal and background classes. Panel (b) is a per-jet score difference, computed under the same class mask as the corresponding subset, and is not derived from panel (a).

Table 5 Final 100k paired SHAP-Delta summary from round 3. Score changes are in probability space; SHAP changes are in XGBoost margin space.

Systematic	Mean Δs, all	Mean Δs, signal	Mean Δs, background	Mean $\|\Delta s\|$, all	Leading feature	Leading mean |ΔSHAP|
esup	0.001164521	-0.002881727	0.00552187	0.02799123	fjet_ECF3	0.1272049
esdown	-0.00805153	-0.01131766	-0.004534003	0.02917145	fjet_ECF3	0.1323322
cpos	3.680653e-05	-4.375854e-05	0.0001235694	0.004166955	fjet_Tau4_wta	0.01281977
cer	-0.0004647173	-0.001544582	0.0006985468	0.02119591	fjet_ECF3	0.06174069

`esup`, ECF_2 has the second-largest response but almost the smallest attribution drift, while L_2 and thrust major, which move by less than 1%, rank third and fourth; under `cpos`, where ECF_3 and τ_4^{WTA} respond almost equally, the drift is led by τ_4^{WTA}. The attribution of an input depends on how much the model relies on it and, through interactions in the trees, on the inputs that do move, so the per-jet response explains which input leads but not the full ordering.

The scale-free test asks the same question of the uncertainty budget (Table 7). Restricted to the nine dimensionless inputs, the same configuration has an energy-scale uncertainty of $2.03 \pm 0.14\%$, against $9.45 \pm 0.11\%$ with all fifteen inputs, a cluster total of 2.75% against 9.91%, and a modelling term of 6.60% against 20.65%, so that the total

Table 6 Per-jet response of the fifteen reconstructed inputs to the four exactly paired calorimeter variations, over all matched eligible jet pairs (4,999,999 for `esup`, `esdown` and `cpos`, 4,995,058 for `cer`; the at most 44 pairs per quantity with a non-finite value are left out). d is the degree of each quantity in the constituent four-momenta. The response of a jet is $r = X^{s}/X^{\mathrm{nom}} - 1$ for the same physical jet. "Uniform" is the response a degree-d quantity would have if every constituent p_{T} were scaled by the median ECF_1 response, $(1.0741)^d - 1$. For `cer` and `cpos`, whose median responses are close to zero, the median absolute response is given instead. All entries are in percent; the full distributions are in Supplementary Table S10.

		Median r [%]			Median $\lvert r\rvert$ [%]	
	d	`esup`	Uniform	`esdown`	`cer`	`cpos`
τ_1^{WTA}	0	−0.94	0	+1.08	2.13	0.023
τ_2^{WTA}	0	−1.45	0	+1.65	2.36	0.055
τ_3^{WTA}	0	−1.73	0	+1.98	2.41	0.090
τ_4^{WTA}	0	−1.87	0	+2.14	2.46	0.115
$\sqrt{d_{12}}$	1	+5.42	+7.41	−5.41	2.67	0.023
$\sqrt{d_{23}}$	1	+4.29	+7.41	−4.30	3.20	0.035
ECF_1	1	+7.41	+7.41	−7.41	2.14	0.001
ECF_2	2	+12.57	+15.36	−11.92	3.59	0.027
ECF_3	3	+17.63	+23.91	−15.94	4.88	0.117
C_2	0	−0.32	0	+0.37	0.72	0.064
D_2	0	+0.28	0	−0.32	1.38	0.049
L_2	0	−0.42	0	+0.47	0.94	0.042
L_3	0	−1.00	0	+1.13	1.40	0.035
Q_w	1	+5.48	+7.41	−5.56	2.27	0.041
Thrust major	0	+0.08	0	−0.09	0.33	0.013

falls from 26.32% to 15.80% (six-seed means and standard deviations). The price is discrimination: the AUC falls from 0.9320 to 0.8761 and the background rejection at 50% signal efficiency from 41.82 to 18.88. Of the four categories only ISR/FSR rises, from 12.22% to 13.81%, almost entirely through the signal final-state radiation envelope, which goes from 11.43% to 13.48%. Adding back the five dimensionful quantities as ratios X/ECF_1^{d}, which restores their shape but not their scale, recovers little: background rejection 19.48 and total uncertainty 16.13%.

Within this feature space, then, the discriminating power and the two largest detector and modelling sensitivities have a common origin: the absolute momentum scale carried by the six dimensionful inputs. With no explicit jet p_{T} or mass among the inputs, these are the only quantities that carry it. They are what the energy-scale variation shifts, and removing them also reduces the generator-modelling term by a factor of three; we have not tested why the generator comparison depends on them. We have not separated how much of that information is jet mass and how much jet p_{T}. The pattern is the trade-off between accuracy and precision that Ref. [11] reports between architectures, seen here inside a single feature space and attributed to specific inputs.

5.5 Conformal coverage

Fig. 4 shows empirical coverage of the class-conditional prediction sets on the nominal test sample and on all nine detector-systematic samples, using the thresholds calibrated once on nominal validation data. Unlike the paired analysis of Section 5.3, this is a dataset-level comparison and needs no pairing, which is why it covers all nine samples. Full numbers are in Supplementary Table S4, and the structure of the prediction sets in Supplementary Table S5.

The class-conditional coverages are the primary result, and they stay close to target everywhere: signal coverage ranges from 89.04% (`esdown`) to 90.12% (`esup`), and background coverage from 89.61% (`esup`) to 90.21% (`esdown`), against the 90% target and against 90.04% and 90.00% on the nominal test sample. Every deviation from target is within one percentage point. The largest single deviation is signal coverage under `esdown`, at 89.04%, followed by signal coverage under `tfj` at 89.58%.

The two energy-scale directions displace the classes in opposite senses, as expected of a two-sided variation: under `esup` signal coverage rises to 90.12% while background coverage falls to 89.61%; under `esdown` signal falls to 89.04% and background rises to 90.21%. `cpos` moves both classes by less than 0.06 percentage points, `bias` by less than 0.03 and `tej` by less than 0.01 — consistent with these being the mildest variations in the background-rejection budget as well.

Overall coverage is reported as a secondary, descriptive number, because it is a mixture of the two class-conditional values weighted by each sample's class composition. On the nominal test sample it is 90.021% against the 90% target, which confirms that the calibration transfers correctly from the validation split to an independent nominal sample. On the varied samples it ranges from 89.60% (`esdown`) to 90.02% (`bias`). For the four track samples the comparison to the nominal overall value is particularly weak: `tfl`, for instance, is 18.5% signal against the nominal 50.5%, so its overall coverage of 89.96% sits much closer to its background-class value than the nominal number

Table 7 The same tagger configuration trained on three input sets, each with six training seeds on the same training jets; entries are the mean and standard deviation over the six seeds, evaluated with the frozen uncertainty procedure on the same test and systematic samples. The nine dimensionless inputs are the $d = 0$ quantities of Table 6. The third set adds the five dimensionful quantities divided by ECF_1^d, which removes their momentum scale but keeps their shape. AUC seed standard deviations are below 5×10^{-5} and are not shown. BR is background rejection at 50% signal efficiency, and every uncertainty is the inclusive relative uncertainty on it, in percent. Per-seed values are in Supplementary Table S9.

Inputs	n	AUC	BR@50%	Energy scale	Cluster	Track	Modelling	ISR/FSR	Total
All fifteen (frozen inputs)	15	0.9320	41.82 ± 0.07	9.45 ± 0.11	9.91 ± 0.11	4.33 ± 0.12	20.65 ± 0.43	12.22 ± 0.37	26.32 ± 0.47
Nine dimensionless ($d = 0$)	9	0.8761	18.88 ± 0.02	2.03 ± 0.14	2.75 ± 0.18	2.79 ± 0.13	6.60 ± 0.22	13.81 ± 0.16	15.80 ± 0.19
Nine + five X/ECF_1^d	14	0.8779	19.48 ± 0.03	2.16 ± 0.08	2.90 ± 0.09	2.49 ± 0.12	6.45 ± 0.14	14.28 ± 0.14	16.13 ± 0.15

does, and the difference between the two overall figures is mostly a statement about class mixture rather than about the tracking variation. No sample produces an empty prediction set, which follows from the overlap of the two acceptance regions noted in Section 4.5, and mean set sizes range from 1.097 to 1.107.

One relation we specifically do not assert is a functional link between the mean score drift of Section 5.3 and the coverage change. Coverage depends on the probability mass near a fixed calibrated threshold, not on the mean of the score distribution, and the two do not track each other here: `esup` shifts the mean signal score down while raising signal coverage. The two diagnostics are reported side by side because they answer different questions, and they are not even defined over the same set of samples.

6 Discussion

6.1 What the results support

Four statements are supported by the frozen artifacts and the extension studies.

The first is about the dataset rather than about any tagger. A robustness study on this release cannot rely on stored high-level branches for the four validated pairable detector variations, or read per-event weights as detector systematics. It has to rebuild the feature space from constituents. We have shown that this can be done to a median relative accuracy of better than 5×10^{-7} per quantity, with 43 of the 44 jets carrying a residual above a 10^{-3} relative tolerance traced to information missing from the released constituents and the remaining one to a reproducible floating-point degeneracy, and we have applied the frozen reconstruction uniformly to 80.5 million jet entries.

The second is that, for this tagger and this feature space, generator modelling dominates the systematic budget. At 50% signal efficiency the modelling term is 20.0% against 10.1% for the calorimeter cluster category and 4.5% for tracking. Removing the two detector categories entirely would take the total from 25.8% to 23.4%, so on this tagger the leverage sits in the generator comparison rather than in the detector terms. This ordering is not a statistical accident: the statistical uncertainty of the modelling term is 0.49% and its spread over training seeds 0.43%.

The third concerns what SHAP and conformal diagnostics add beyond a performance number. They are sensitive to different things. Attribution drift under the energy-scale variations is large per jet, concentrated on the energy correlators, and yet barely moves the importance ordering; conformal coverage under the same variations moves by less than a percentage point but does so in opposite directions for the two classes. Neither of these is visible in AUC, which changes by 0.003 between nominal and `esup`. We regard both as diagnostics that describe how a classifier's behaviour shifts, not as significance tests.

The fourth concerns where this tagger's sensitivity comes from. The inputs that respond most to the calorimeter energy scale are those of highest degree in the constituent momenta, the largest attribution drift falls on the one of highest degree, and removing the absolute momentum scale from the inputs removes most of the energy-scale and modelling uncertainty together with more than half of the background rejection. These are measurements on one model class and one feature space. The degree of each input, however, is a property of its definition rather than of gradient boosting, so the same input-level reasoning applies to any tagger built on these quantities, although we have not tested how other architectures weigh them.

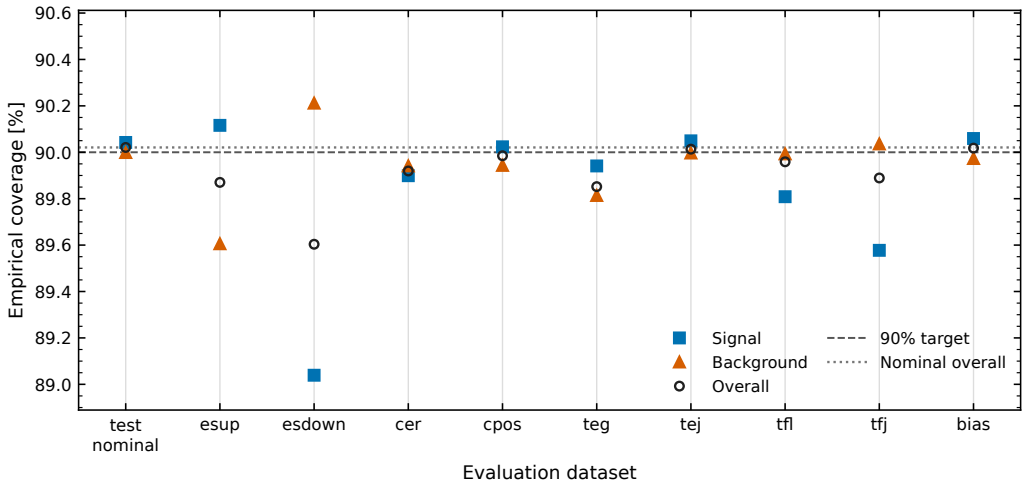


Fig. 4 Empirical coverage of class-conditional split-conformal prediction sets on the nominal test sample and the nine detector-systematic samples. Thresholds were calibrated once on the frozen nominal validation split at a 90% target (dashed line) and applied unchanged everywhere; the dotted line marks the overall coverage measured on the nominal test sample. The horizontal axis is categorical and carries no ordering, so the markers are drawn unconnected against light vertical guides. The two class-conditional coverages are the primary quantities; the overall value is a mixture of them weighted by each sample's class composition, which differs strongly between the calorimeter and the track samples. Full values are given in Supplementary Table S4.

6.2 Limitations

The training sample is the clearest limitation. The classifier was developed from a five-million-jet nominal subset taken from the first fifty released `train_nominal` files, of which 4,499,612 jets were used for fitting and 500,385 held out. The released training partition is much larger, and the files were taken as a contiguous block rather than a random draw across the partition; we did not demonstrate that this block is representative of the whole, and we did not run a training-size scan. The size is close to the six million jets of the reference training script, but the dataset documentation states that state-of-the-art performance requires the entire partition [30]. The gap to the published hlDNN reference in Table 3 is therefore not attributable to any single cause, and we do not attempt to decompose it.

The two diagnostics have different scope, and neither should be read as covering the other's samples. Paired SHAP-Delta covers four of the nine detector systematics, the calorimeter-cluster variations that pair jet-by-jet; that restriction is forced by the data, as Section 4.4 and Table 2 document, and it means the attribution results speak only to those four. Conformal coverage covers all ten evaluated samples, but it does so at the level of whole datasets rather than matched jets, so a coverage difference between a varied sample and nominal is a difference between two populations and not a per-jet effect. Extending the *paired* diagnostics to the tracking systematics would require a release in which those samples are re-simulations of the same jets.

The SHAP results are for one model. TreeSHAP is exact for the fitted trees, so there is no attribution-estimator uncertainty, but the attributions themselves depend on the training sample, the seed and the hyperparameters, and we quantify only the dependence on the evaluation sample size; the six-seed study of Section 4.2 covers performance and uncertainties, not attributions. Carrying the same paired construction to a second architecture would test how much of the pattern is a property of the feature space rather than of gradient boosting.

The conformal results are diagnostic, not inferential. Coverage was measured under distribution shift with thresholds calibrated on nominal data, a setting in which the conformal guarantee does not formally hold [46]. We report empirical coverage and its deviation; we do not attach a confidence statement to it, and the deviations we observe are small enough that we would not read a physical claim into their ordering.

The statistical uncertainties of this paper describe the finite size of the evaluation samples. The six-seed study varies the training seed but not the training jets, so the statistical uncertainty of the training sample itself is not covered, and a comparison across disjoint training blocks remains to be done. Because each uncertainty component is a modulus, its bootstrap distribution is skewed when the component is small, and the standard deviations quoted for `cpos`, track bias and the smaller binned entries indicate a size rather than a Gaussian error. The input-set variants have only the six-seed spread and no bootstrap of their own. Their differences from the full tagger are many times the statistical precision of the frozen result, but the small difference between the uncertainties of the two reduced input sets should not be interpreted.

Finally, the binned uncertainty of Fig. 2(b) covers 350–3150 GeV, the range the reference implementation defines. That range holds 8,959,008 of the 10,306,812 eligible test jets, or 86.9%; the remaining 13.1% enter the inclusive numbers but no p_{T} bin, and they are signal-enriched (61.8% signal, against 48.8% inside the binning and 50.5% overall), so the binned and inclusive numbers are not measurements of the same jet population. The 1.35–1.55 TeV interval carries the official extrapolation convention rather than a measurement of its own.

7 Conclusion

We have rebuilt the fifteen-variable high-level feature space of the ATLAS top-tagging open dataset from the released jet constituents. For the four exactly pairable calorimeter variations, the stored high-level branches remain nominal despite constituent changes; the per-event weight array holds theory rather than detector variations. The reconstruction closes against the stored nominal branches to a median relative accuracy between 2.5×10^{-8} and 4.8×10^{-7} per quantity, with 0.072% of quantity–jet comparisons outside a 10^{-3} tolerance. Those residuals fall on 44 jets, of which 43 are information-loss cases and one is a reproducible floating-point degeneracy; no proven formula or implementation mismatch was found and no jet was left unresolved. The reconstruction was frozen and applied to 80,481,421 jet entries across seventeen samples.

An XGBoost tagger developed from a five-million-jet nominal subset — 4,499,612 jets for fitting, 500,385 held out — reaches an unweighted AUC of 0.9320, accuracy 0.8559, and background rejection 41.91 at 50% signal efficiency on 10,306,812 test jets. Evaluated with the procedure the record prescribes, the total relative systematic uncertainty on background rejection is 25.8%: generator modelling 20.0%, ISR/FSR scales 12.0%, calorimeter clusters 10.1%, tracking 4.5%. A bootstrap that preserves the jet-by-jet pairing of the calorimeter variations gives the total a statistical uncertainty of 0.6 percentage points, and six training seeds spread it by 0.5; every inclusive entry of the uncertainty table except `cpos` and track bias is at least five statistical standard deviations from zero, and the p_{T} dependence of the energy-scale and cluster terms is genuine while that of the track terms is compatible with statistics. The published ATLAS high-level network is quoted alongside only as an external reference in the same feature space.

On the four calorimeter-cluster variations that pair jet-by-jet with the nominal sample, paired TreeSHAP attribution drift concentrates on ECF_3 for the energy-scale and energy-resolution variations, followed by ECF_1 and L_2 under the energy scale and by τ_4^{WTA} under the energy resolution, and on τ_4^{WTA} at an order of magnitude smaller size for the position variation, while global importance rankings change minimally: the Spearman correlation between nominal and varied orderings is at least 0.993 for all-jet rankings and at least 0.986 across every systematic-by-class combination. Class-conditional conformal prediction sets calibrated once on nominal validation data and applied unchanged to the nominal test sample and all nine detector-systematic samples retain class-conditional coverage within one percentage point of the 90% target throughout, with the two energy-scale directions displacing the signal and background classes in opposite senses.

The leading attribution drift follows the degree of each input in the constituent momenta: under `esup` ECF_3, of degree three, moves by a median 17.6% per jet, against 7.4% for the summed constituent momentum and less than 2% for every dimensionless input. Retraining on the nine dimensionless inputs lowers the six-seed mean total uncertainty from 26.3% to 15.8%, through the energy-scale and modelling terms, but lowers background rejection from 41.8 to 18.9, so within this

feature space the discrimination and the leading sensitivities share one origin, the absolute momentum scale of the dimensionful inputs.

The reading we would defend is narrow. A tagger's response to a systematic variation has structure that a single performance number does not show, and on a public dataset that structure can be measured jet-by-jet once the feature space is reconstructed correctly.

Statements and declarations

Data Availability Statement

This manuscript has associated data in a data repository. [Authors' comment: This study analyses the ATLAS top tagging open data set with systematic uncertainties, CERN Open Data Portal Record 80030, https://doi.org/10.7483/OPENDATA.ATLAS.SOAY.LABE.]

Code Availability Statement

This manuscript has associated code/software in a data repository. [Authors' comment: The analysis code, the trained tagger and the frozen results quoted in this paper are publicly available at https://github.com/adeelrahmat64-beep/atlas80030-tagger-robustness. Release v1.1.0 adds the scripts and outputs of the statistical-precision, training-seed and input-degree studies (`uncertainty_chain.py`, `stat_bootstrap.py`, `tagger_variants.py` and `feature_response.py`) to the first release, v1.0.0 (https://doi.org/10.5281/zenodo.23083866). Every release is archived on Zenodo under the concept DOI https://doi.org/10.5281/zenodo.23083865, which always resolves to the latest version.]

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